MAHATHMA GANDHI UNIVERISTY

BCA (HONOURS)SECOND SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG2CCRBCA101– Data Structures

Model Question Paper

Duration: 2hrs Maximum Marks: 70

*Students should attempt at least one question from each course outcome to enhance their overall outcome*attainability.

Part A

Very Short Answer Questions

Answer All Questions

Each question carries 2 marks

1.Differentiate between linear and non-linear Data Structures. [U][CO1]

2.How does the pop operation change the state of the stack in an array-based implementation? [U][CO2]

3.Define a linked list and explain its basic structure. [U][CO3]

4.What is a walk in a graph? How does it differ from a path? [An][CO4]

5.Discuss the disadvantages of Binary Search? [A][CO1]

(5*2=10 Marks)

Part B

Short Answer Questions (5 out of 7 Questions)

Each question carries 6 marks

6.Differentiate between best-case and worst-case complexities. [An][CO1]

7.Compare the advantages and disadvantages of using postfix notation over infix notation for arithmetic expression evaluation. [A][CO2]

8. Explain with diagrams how a circular linked list is implemented. What are its advantages over a singly linked list? [A][CO3]

9.Compare recursive implementation and iterative implementation with suitable examples. [An][CO3]

10.Discuss the similarities and differences of Insertion Sort and Selection Sort. [An][CO1]

11. Explain the array implementation of a simple queue and circular queue with algorithms and examples. [A][CO2]

12. Explain insertion in an AVL tree with a suitable example. [U][CO4]

(5*6=30 Marks)

Part C

Essay Questions (2 out of 3 Questions)

Each question carries 15 marks

13.Explainthe mergingof two sorted arrays with an algorithm and a suitable example. [A][CO1]

14.Compare and contrast circular queue, Double-ended-queue and priority queues with examples. [An][CO2]

15.Explain hashing. Discuss various collision handling methods that can be implemented to reduce collision. [U][CO3]

(2*15=30 Marks)